B-lymphocytes (B cells) or Plasma cells in response to exposure to antigen. Antibodies are also known as immunoglobulins. They react specially with that antigen in vivo or in vitro and are hence a part of the adaptive immune response specifically, humoral immunity.

13.6.1 Structure of an Immunoglobulin

1. Basic unit

The basic structural unit (monomer) of an immunoglobulin molecule consists of four polypeptide chains linked covalently by disulfide bonds (Figure 13.21). The four-chain structure is composed of two identical light (L) and two identical heavy (H) polypeptide chains. Every immunoglobulin can be represented by the general formula $(H_2L_2)_n$.

a) Light chains

Light Chains have a molecular weight of approximately 25000 Da and are composed of about 220 amino acids. Light chains are common to all immunoglobulin classes and are of two types – kappa (κ) or lambda (λ) - based on their structural differences. A given immunoglobulin molecule may contain either identical κ or λ chains but never both.

b) Heavy chains

Heavy chains have a molecular weight of approximately twice that of light chains (57000-70000 Da) and twice the number of amino acids (about 440). Five antigenically distinct isotypes of heavy chains are recognized-gamma (γ), alpha (α), mu (μ), delta (δ) and epsilon (ε) – based on structural differences in the carboxy terminal portion of heavy chains. The heavy chains isotypes form the basis of five classes of immunoglobulin molecules – IgG (contains γ chain), IgA (contains α chain),

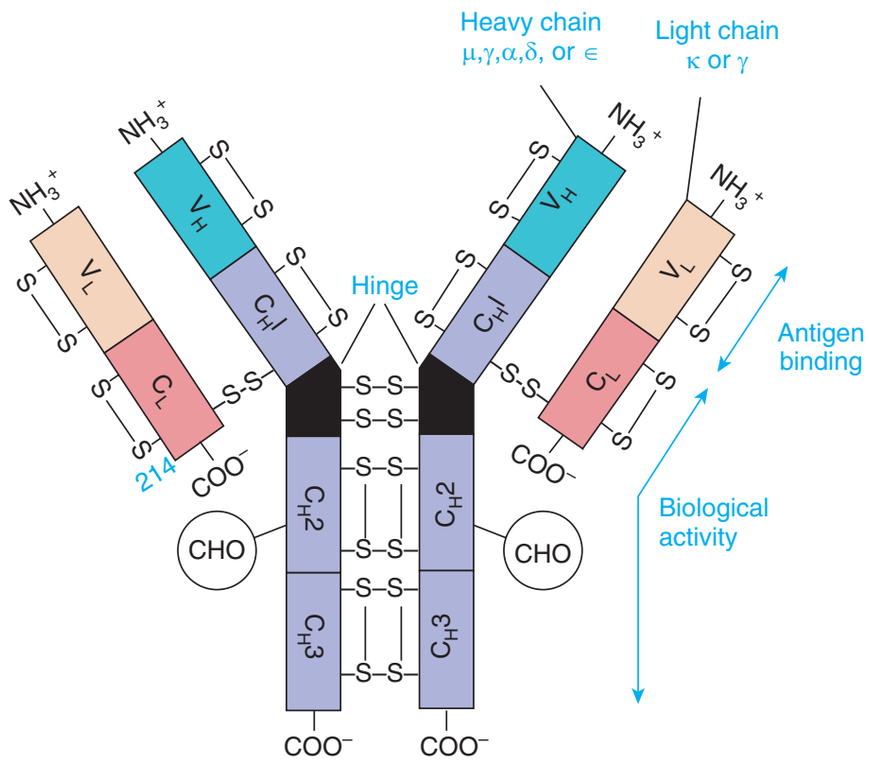

Figure 13.25: Structure of Immunoglobulin

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